GENERATIVE AI · END-SEMESTER MINI PROJECT

tarangAI

Multimodal sentiment analysis for English and Malayalam-English speech: reading what was said and how it was said

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Source code: https://github.com/adi5022/tarangAI

Models and services: Sarvam AI (Saaras, chat model) · Groq-hosted Whisper · audEERING wav2vec2 (Hugging Face)

Abstract

People increasingly send voice notes in a mix of English and Malayalam, yet most sentiment tools assume clean English text. They also ignore a second signal that humans use constantly: the voice itself. **tarangAI** is a Telegram bot that accepts a voice note or a typed message, transcribes it with a language-aware speech pipeline, and returns a structured reading of the speaker's state: sentiment (positive, negative or neutral), a main emotion, an intensity level and a sarcasm flag.

The pipeline routes English speech to Whisper and Malayalam or code-mixed speech to Sarvam Saaras, and runs sentiment analysis with a Sarvam chat model *directly on the original text*, with no translation step that could flatten slang or tone. For English voice notes, a pretrained Hugging Face speech-emotion model additionally estimates the *arousal* and *valence* of the voice. A small, transparent fusion layer uses these to refine intensity and to flag cases where the words and the voice disagree, such as cheerful delivery over negative words, a common sign of sarcasm. The voice cut-offs were calibrated on 25 recorded clips rather than guessed.

Live tests showed correct transcripts, sentiment, emotion and sarcasm on four English voice notes, an end-to-end latency of about 1.3 to 2.4 seconds per note, and a useful disagreement flag on a sarcastic sentence. The project also documents its limits: the voice model is English-trained and calibrated on one speaker, and no formal accuracy evaluation has been run yet.

1. Motivation

Language reality. Malayalam speakers routinely switch between Malayalam and English inside one sentence, and write Malayalam in Latin letters ("Manglish"): "*movie poli aanu, climax mass*". Models built for formal English handle this poorly, and voice notes make it harder still, because speech recognition for code-mixed speech is a separate problem.

Words are only half the message. The sentence "I'm fine, everything is great" can be sincere, tired or bitterly sarcastic. A text-only classifier sees identical input in all three cases and must return the same answer. Figure 4 shows this happening in our own tests. People resolve the ambiguity from tone of voice: pitch, energy and pace. A system that only reads the transcript throws that away.

Why this fits a generative-AI project. The project composes several foundation models (speech-to-text, a large language model and a self-supervised speech encoder) into one working system, and the interesting engineering is in the glue: routing by language, constraining model output to a closed vocabulary, and fusing two modalities without letting either one silently override the other.

2. Problem statement

Build a system that, given a voice note or message in English or Malayalam-English, returns a reliable and *inspectable* estimate of sentiment, emotion, intensity and sarcasm; uses vocal cues where a suitable model exists; and exposes every decision (transcript, language, adjustments) to the user so errors can be traced.

- **Inputs:** Telegram voice notes (OGG/Opus) and typed text in English, Malayalam script or Manglish.
- **Outputs:** transcript, language, sentiment, emotion, intensity, sarcasm flag and, for English voice, voice arousal/valence with any adjustment made.
- **Constraints:** low latency (a few seconds), low cost (free-tier APIs plus one local model), no translation step, and graceful degradation when a component is unavailable.

3. Use cases

Use case	How tarangAI helps
Customer feedback in Kerala	Businesses receive voice and text complaints and praise in mixed Malayalam-English. Structured sentiment and emotion make triage and trend counting possible without reading every note.
Support-call and voice-note triage	The intensity and the "words vs voice" flag surface calls where the customer says "okay" in a clearly frustrated voice.
Sarcasm-aware moderation	Cheerful delivery over negative wording is flagged, which text-only filters miss.
Student and course feedback	Short spoken feedback from students who are more comfortable speaking than typing, in their own mix of languages.
Accessibility	People who prefer speaking, or who write Malayalam in Latin script, get the same analysis quality as typists in English.
Research baseline	A reusable, transparent pipeline for experiments on code-mixed sentiment and text-plus-voice fusion.

These are intended applications for the prototype. tarangAI has not been deployed or validated for any of them, and it is not a clinical or diagnostic tool (see Section 9).

4. System design

Every incoming message passes through one routing function. The central design rule is that **voice analysis runs only on the English (Whisper) path**, because the voice model was trained on English speech. Malayalam and code-mixed audio, and typed text, skip it.

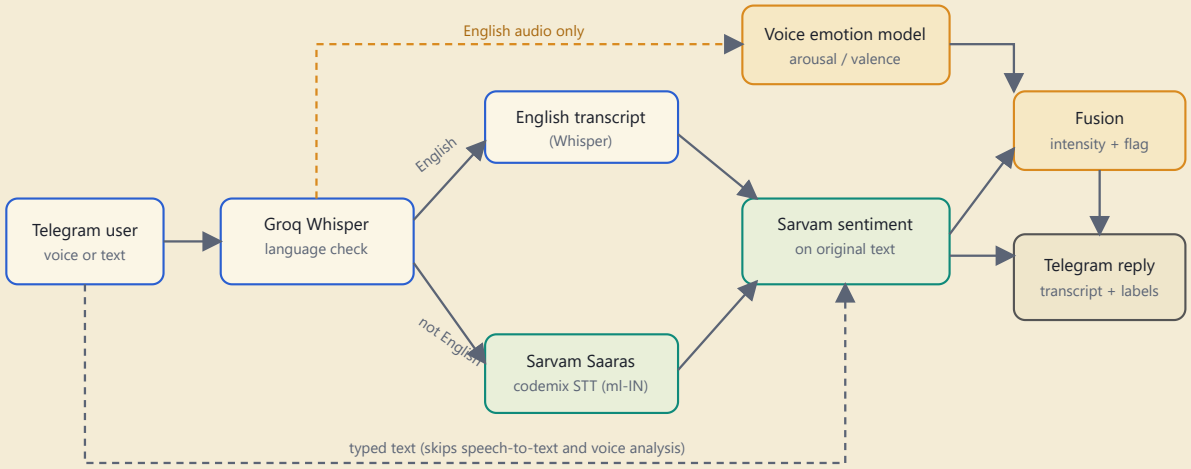


Figure 1. System architecture. Solid arrows: voice. Dashed orange: audio sent to the voice model, English only. Dashed grey: typed text.

4.1 Components

Stage	Choice	Why
English speech-to-text and language check	Groq-hosted `whisper-large-v3`	Fast, free-tier friendly, accurate on English. Used only for "is this English?" and English transcripts.
Malayalam / code-mixed speech-to-text	Sarvam `saaras:v3`, codemix mode, `ml-IN`	Built for Indian languages. Whisper is never trusted for Malayalam: it often mislabels it as another language.
Sentiment, emotion, intensity, sarcasm	Sarvam `sarvam-105b-conversations`	One short call returning a constrained line such as `negative|anger|strong|no`.
Voice emotion (English only)	`audeering/wav2vec2-large-robust-12-ft-emotion-msp-dim`	Outputs arousal, valence and dominance from raw audio. Dimensional output transfers better than fixed emotion words.
Interface	Telegram bot (python-telegram-bot)	Voice notes are native to Telegram; no app to build.

Table 1. Components and rationale. The speech-to-text routing was adapted from Sahayi, an earlier project by Adithyan RS and collaborators (reference 7).

4.2 Output fields

Field	Values	Source
Sentiment	positive · negative · neutral	Sarvam chat model
Emotion	joy, love, surprise, sadness, anger, fear, disgust, none	Sarvam chat model
Intensity	mild · moderate · strong	Sarvam, then possibly adjusted by voice
Sarcasm	yes · no	Sarvam chat model
Voice arousal / valence	about 0 to 1, shown as low / mid / high	Hugging Face voice model (English audio only)

Table 2. Every field comes from a closed list, so outputs are comparable between runs and cannot drift into free text.

5. Methodology

5.1 Language-aware speech routing

A single Whisper call listens first. If it reports English, its transcript is used and Sarvam is never called. Anything else goes to Sarvam Saaras in code-mix mode. Audio longer than 28 seconds uses Sarvam's batch API, because the real-time endpoint has a 30-second limit. Voice notes arrive as OGG/Opus and are handled without writing the audio to disk.

5.2 Sentiment on the original text

The first design translated code-mixed text to English before classifying it. We dropped that step: translation costs an extra model call and can erase exactly the slang, tone and sarcasm the classifier needs. The prompt tells the model that the input may be English, Malayalam or a mixture in native or Latin script, and asks for one line of four fields separated by `|`. The reply is parsed against closed lists, with safe defaults for any field that fails to parse; if no valid sentiment is found, the step reports an error instead of guessing.

5.3 Voice emotion model

The audEERING wav2vec2 model (references 2 and 3), fine-tuned on the English MSP-Podcast corpus (reference 4), takes a 16 kHz waveform and regresses three dimensions in roughly the range 0 to 1: **arousal** (calm to energetic), **valence** (negative to positive) and **dominance**. Clips under 1 second are rejected as unreliable, and clips over 30 seconds are cut to the first 30. The model loads once at startup. If it fails to load, voice analysis switches itself off and the rest of the bot keeps working.

5.4 Fusion rules

The text labels are the source of truth and are **never changed** by voice. Voice may only do two things, and both are shown to the user.

Condition	Effect
Voice arousal high	Raise text intensity by one step (mild → moderate → strong)
Voice arousal low	Lower text intensity by one step
Text positive, voice valence low	Flag: "Words sound positive but the voice sounds negative"
Text negative, voice valence high	Flag: "Words sound negative but the voice sounds positive"

Table 3. Fusion rules. The cut-offs for "low" and "high" come from the calibration in Section 6.

6. Calibration of the voice cut-offs

Raw arousal and valence are continuous scores, and what counts as "high" depends on the speaker and microphone. Rather than guess thresholds, we built a small Streamlit tool with a guided 25-clip script. Part A is neutral sentences said three ways (calm, normal, loud and urgent). Part B is negative and positive sentences said with a clearly unhappy or happy tone. The tool averages the clips for each intended level, places each cut-off halfway between neighbouring averages, and refuses to suggest values if the class means are out of order.

Dimension	Low mean (n=5)	Mid mean (n=15)	High mean (n=5)	Cut-offs set (low / high)
Arousal	0.275 (sd 0.054)	0.471 (sd 0.122)	0.735 (sd 0.076)	**0.373 / 0.603**
Valence	0.282 (sd 0.079)	0.401 (sd 0.081)	0.714 (sd 0.162)	**0.342 / 0.557**

Table 4. Calibration results: 25 clips from one speaker, one microphone. Means are in the expected order for both dimensions.

> **Caveat.** The mid-arousal clips vary widely (sd 0.12), so labels near the cut-offs are noisy. The values fit one voice and one microphone; another speaker may land elsewhere on the scale. Voice scores are therefore treated as an indicative second signal, not a verdict.

7. Experiments and results

All experiments below are small, manual checks. They show that components work and expose failure modes; they are **not** a statistical evaluation, and no accuracy percentage is claimed.

7.1 Live test: English voice notes in Telegram

The author recorded five English voice notes in Telegram and sent them to the bot. Figures 2 and 3 show the replies.

What was said	Sentiment · emotion	Intensity	Arousal	Valence	Fusion outcome
"I'm so happy with how today turned out."	positive · joy	moderate	0.57 mid	0.60 high	no change
"I can't believe they cancelled my order again."	negative · anger	strong	0.72 high	0.27 low	no change (already strong)
"I miss my family so much right now."	negative · sadness	strong → moderate	0.34 low	0.46 mid	intensity lowered: calm voice
"Oh great, another delay. Just what I needed." (sarcastic)	negative · anger · sarcasm	strong	0.71 high	0.69 high	**flag: words negative, voice positive**
"The meeting is at 5."	neutral · none	mild → moderate	0.67 high	0.45 mid	intensity raised: energetic voice

Table 5. Live results. Transcripts were exact in all five. End-to-end time per note was 1.3 to 2.4 seconds (from the bot's logs).

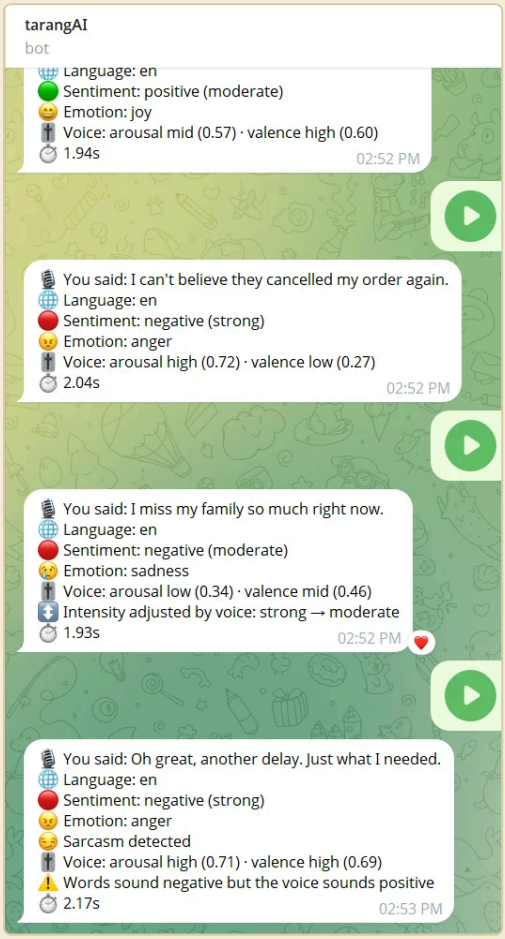


Figure 2. Replies to the happy, angry, sad and sarcastic notes. Note "intensity adjusted by voice" on the sad clip and the "words and voice disagree" flag on the sarcastic one.

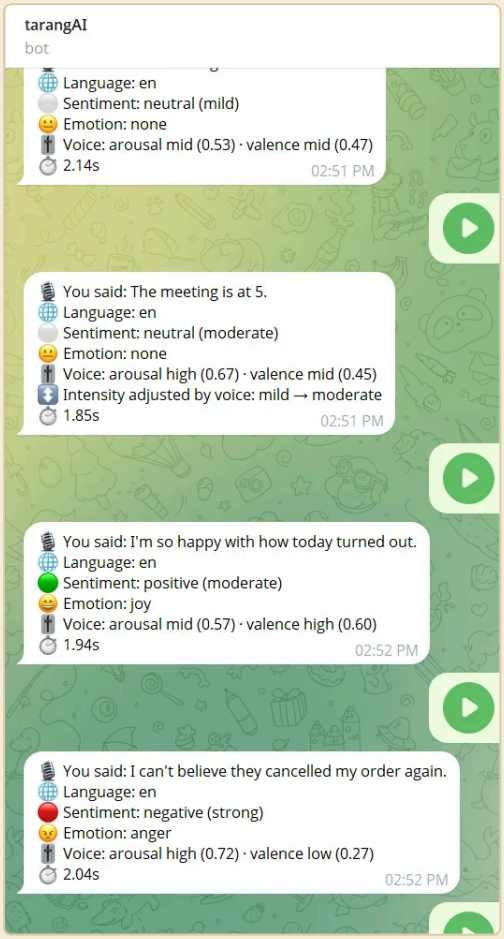


Figure 3. Neutral clip with intensity raised by an energetic voice (mild to moderate), followed by the happy and angry notes.

7.2 The sarcasm case

The sarcastic note is the clearest demonstration of the fusion idea. The language model read the words correctly as negative, anger and sarcasm. The voice model independently reported a *positive-sounding* voice (valence 0.69), so the bot added the disagreement flag. Neither signal alone says "sarcasm" as convincingly as the pair does. This is one example from one speaker, not evidence of general sarcasm detection.

7.3 Speech-to-text paths agree on clean speech

To check that the two STT paths behave consistently, we generated three English clips with Windows text-to-speech, encoded them as Telegram-style Opus, and sent each through both the Sarvam path and the Whisper path.

Clip	Sarvam transcript	Whisper transcript	Sentiment (both)
happy	"I am so happy with how today turned out. This is wonderful news. Thank you."	"I am so happy with how today turned out. This is wonderful news, thank you."	positive · joy · strong
neutral	"The meeting is at 5:00 in the main hall."	"The meeting is at 5 o'clock in the main hall."	neutral · none · mild
angry	"I can't believe they cancelled my order again. This is the worst service ever."	(identical)	negative · anger · strong

Table 6. Both paths produce equivalent transcripts and identical sentiment on clean synthetic speech. Real, noisy, accented speech was not compared this way.

7.4 Typed text: Malayalam script, Manglish and English

Eight typed samples were sent straight to the sentiment step. All outputs parsed and were plausible.

Input	Sentiment	Emotion	Intensity	Sarcasm
My day has been good so far.	positive	joy	mild	no
movie poli aanu, climax mass	positive	joy	moderate	no
ഇന്ന് ഒരു മോശം ദിവസമായിരുന്നു	negative	sadness	strong	no
wow great, the bus broke down again, thanks a lot	negative	anger	strong	**yes**
this is the worst bus service ever	negative	anger	strong	no
ഇന്ന് ക്ലാസ് ഉണ്ട്	neutral	none	mild	no
enthina ee padam kandathu, time poyi	negative	anger	moderate	no
ammaye orth ipol karachil varunnu	negative	sadness	strong	no

Table 7. Multi-field output on English, Malayalam script and Manglish. Expected labels were judged by the author, not taken from a labelled dataset.

7.5 Robustness to short or ambiguous input

Voice notes sometimes yield very short or odd transcripts. We tested twelve such inputs to check that the classifier does not fall back on a negative default (our prompt contains one worked example, which can bias a model). It did not: "hello", "ok", "hmm", "testing testing one two three", "you" and "I don't

know what to say" all returned *neutral*; "I just got the job offer!" and "Thank you so much, that really helped." returned *positive · joy*; "I'm bored." returned *negative · sadness · mild*. Removing the example from the prompt gave the same sentiment on every input, with only small changes in emotion and intensity.

7.6 What text-only analysis cannot do



Figure 4. The same sentence, "I'm fine, everything is great", sent several times in a row. Without voice analysis the bot returns the same reading each time (negative · sadness · sarcasm), differing only in intensity.

This screenshot was taken before the voice step was active for the author's English notes, so it is effectively the **text-only baseline**. The transcripts are all correct; the sentiment is the same because the words are the same. It illustrates the motivation directly: when the words do not change, a text classifier cannot see tone. The intensity wobbling between mild, moderate and strong across repeats also shows that intensity from a language model alone is unstable, one reason the voice cue is used to adjust it.

8. Challenges and lessons

- **Silent fallback hid a configuration error.** English voice notes were being routed to Sarvam because the Groq key was set under a different variable name (`GROQ_API_KEY_3`) and the Whisper failure was swallowed. The symptom was that the voice-analysis lines never appeared. Fixes: accept all three key names with failover, and log a warning when the Whisper check fails.

- **Model names and parameters change.** The first Sarvam model, `sarvam-m`, has been deprecated, and the documentation summary suggested a reasoning setting (`none`) that the live API rejects with a 400 error. Probing the real endpoint showed that the larger model also spent small token budgets on hidden reasoning and returned empty answers; the `-conversations` variant answers directly. Lesson: test the live API, do not rely on documentation alone.
- **Library drift.** The voice model's published example code failed under transformers 5.x until a single call was changed to `post_init()`.
- **Translation was the wrong abstraction.** An early design normalised code-mixed text to English first. Removing it simplified the pipeline and avoided losing tone.
- **Debuggability.** An opt-in debug log (`DEBUG_TRANSCRIPTS=1`) prints transcripts, labels and voice scores. It is off by default because transcripts are private messages.

9. Limitations and ethics

Limitations

- **No formal evaluation.** Results come from manual spot checks. There is no labelled test set, so no accuracy or F1 is reported.
- **The voice model is English-trained** (MSP-Podcast) and is deliberately not applied to Malayalam or code-mixed audio, where its behaviour is unknown.
- **Calibration covers one speaker and one microphone** with 25 clips. Other voices, accents and noisy rooms may shift the scores.
- **Intensity and sarcasm are the least reliable fields**; the sentiment label is the most dependable output.
- **Language coverage.** Sarvam STT is configured for Malayalam (`ml-IN`), so Hindi-English speech is not supported.
- **Language-model subjectivity.** Sentiment from an LLM can disagree with human annotators, especially on mixed feelings.

Ethics and privacy

- **Not a clinical tool.** The bot estimates tone in a message; it does not assess mental health, and its outputs must not be used for diagnosis or for decisions about people.
- **Private data.** Voice notes and transcripts are personal. The bot does not store audio or transcripts, and logs them only when debugging is switched on. Audio and text are however processed by third-party services (Telegram, Groq, Sarvam) under their own policies, so users should be told this and consent.
- **Transparency.** Every adjustment made by voice is shown in the reply, and the text labels are never silently overridden.
- **Bias.** Emotion models trained on acted or English podcast speech can under-perform for some accents and speakers. This is a real risk for Malayalam-accented English.
- **Licence.** The voice model is released under CC BY-NC-SA 4.0 for research use. This project is therefore non-commercial unless the model is replaced or licensed.

10. Future work and conclusion

Future work

- Run a **labelled evaluation**: score the sentiment field (macro-F1 and a confusion matrix) on a public code-mixed Malayalam-English sentiment dataset such as Dravidian-CodeMix (reference 5; Roman-script YouTube comments, about 4,850 training and 1,350 test comments in the FIRE 2020 release). Its label set includes "mixed feelings" and "unknown" classes, so a mapping to our three classes is needed. The data is text only, so it would test the text path, not the voice path. Compare raw text against a translated baseline too.
- Evaluate on **real, noisy, multi-speaker** recordings, and calibrate per user with a short baseline.
- Find or fine-tune a **voice-emotion model for Malayalam** and code-mixed speech so voice cues are available on every path.
- Replace the model's self-reported confidence with token probabilities where the API exposes them.
- Add opt-in conversation history so trends over time, not just single messages, can be reported.

Conclusion

tarangAI shows that a small, inspectable pipeline can combine speech recognition, a language model and a speech-emotion model into a useful reading of code-mixed voice notes. Its main design choices (language-aware routing, no translation, closed-vocabulary outputs, and a fusion layer that adjusts intensity and flags disagreements without overriding the words) all came from problems met while building it. Live tests confirmed correct transcripts and labels on English voice notes and a working words-versus-voice flag on a sarcastic sentence. The honest next step is a labelled evaluation, which would turn these promising demonstrations into measured results.

References and resources

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8. python-telegram-bot, Streamlit, PyTorch and Hugging Face Transformers documentation.
9. tarangAI source code: github.com/adi5022/tarangAI

Bibliographic details were checked against online sources (arXiv, PMLR, the Hugging Face model card and the shared-task page) in November 2026. Volume and page numbers are omitted where they could not be confirmed.